PROGRAM

Import Libraries

```
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
from sklearn.model_selection import train_test_split
from sklearn.ensemble import RandomForestClassifier
from sklearn.metrics import accuracy_score
from sklearn.metrics import confusion_matrix, ConfusionMatrixDisplay
```

Upload Dataset

```
from google.colab import files
uploaded = files.upload()
data = pd.read_csv("food_calorie_dataset.csv")
print(data.head())
print("\nDataset shape:", data.shape)
```

OUTPUT

	Food_Name	Calories_per_100g_kcal
0	Idli	58
1	Dosa	168
2	Plain Rice	130
3	Biryani	180
4	Chapati	120

Dataset shape: (21, 2)

```
print(data)
```

OUTPUT

	Food_Name	Calories_per_100g_kcal
0	Idli	58
1	Dosa	168
2	Plain Rice	130
3	Biryani	180
4	Chapati	120
5	Poori	330
6	Vada	290
7	Pongal	160
8	Sambar	75
9	Curd Rice	150
10	Lemon Rice	175
11	Tomato Rice	170
12	Vegetable Rice	160
13	Chicken Biryani	200
14	Chicken Curry	190
15	Fish Curry	150
16	Paneer Butter Masala	180
17	Potato	77
18	Banana	89
19	Apple	52
20	chutney	100

Prepare The ML Data

```
from sklearn.model_selection import train_test_split
from sklearn.ensemble import RandomForestClassifier

X = data[["Calories_per_100g_kcal"]]
y = data["Food_Name"]

print("Input:")
print(X.head())
print("\nTarget:")
print(y.head())
```

OUTPUT

Input:

	Calories_per_100g_kcal
0	58
1	168
2	130
3	180
4	120

Target:

0	Idli

```
1          Dosa
2    Plain Rice
3       Biryani
4       Chapati
Name: Food_Name, dtype: object
```

Train Random Forest

```
model = RandomForestClassifier(
    n_estimators=100,
    random_state=42
)

model.fit(X, y)
print("Random Forest model trained successfully!")
```

OUTPUT

```
Random Forest model trained successfully!
```

Test The Model

```
y_pred = model.predict(X)

print("Predicted Food Names:")
for actual, predicted in zip(y, y_pred):
    print("Actual:", actual, " | Predicted:", predicted)
```

OUTPUT

```
Predicted Food Names:
Actual: Idli  | Predicted: Idli
Actual: Dosa  | Predicted: Dosa
Actual: Plain Rice  | Predicted: Plain Rice
Actual: Biryani  | Predicted: Paneer Butter Masala
Actual: Chapati  | Predicted: Chapati
Actual: Poori  | Predicted: Poori
Actual: Vada  | Predicted: Vada
Actual: Pongal  | Predicted: Pongal
Actual: Sambar  | Predicted: Sambar
Actual: Curd Rice  | Predicted: Fish Curry
Actual: Lemon Rice  | Predicted: Lemon Rice
Actual: Tomato Rice  | Predicted: Tomato Rice
Actual: Vegetable Rice  | Predicted: Pongal
Actual: Chicken Biryani  | Predicted: Chicken Biryani
Actual: Chicken Curry  | Predicted: Chicken Curry
Actual: Fish Curry  | Predicted: Fish Curry
Actual: Paneer Butter Masala  | Predicted: Paneer Butter Masala
Actual: Potato  | Predicted: Potato
Actual: Banana  | Predicted: Banana
Actual: Apple  | Predicted: Apple
Actual: chutney  | Predicted: chutney
```

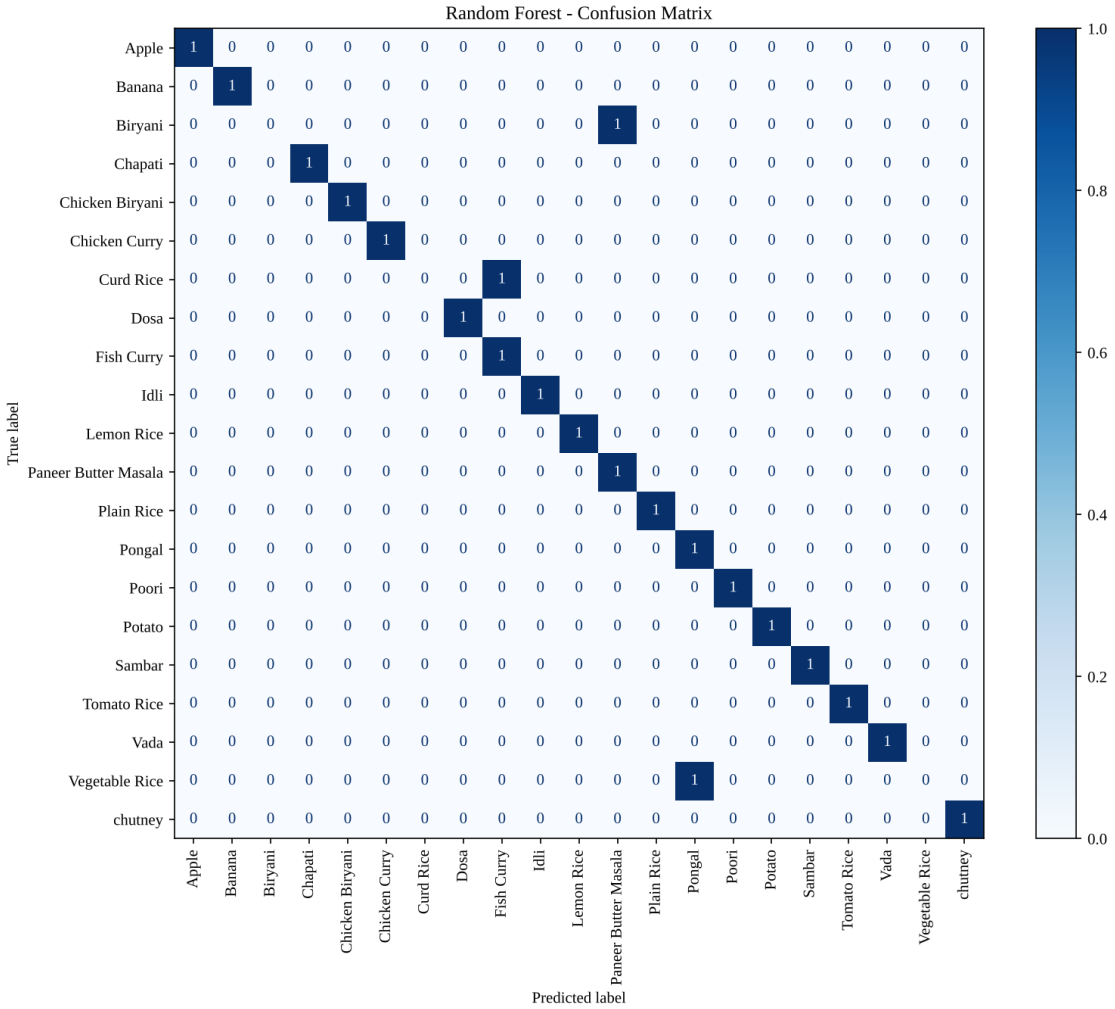
Confusion Matrix

```
from sklearn.metrics import confusion_matrix, ConfusionMatrixDisplay
import matplotlib.pyplot as plt

cm = confusion_matrix(y, y_pred, labels=model.classes_)
disp = ConfusionMatrixDisplay(
    confusion_matrix=cm,
    display_labels=model.classes_
)

fig, ax = plt.subplots(figsize=(12, 10))
disp.plot(
    ax=ax,
    xticks_rotation=90,
    cmap="Blues"
)
plt.title("Random Forest - Confusion Matrix")
plt.show()
```

OUTPUT



Upload Food Image

```
uploaded = files.upload()
image_name = list(uploaded.keys())[0]
print("Image uploaded:", image_name)
```

Calorie Calculation

```
foods = ["Dosa", "Sambar", "Chutney"]
quantities = [150, 100, 50]
calories = []

for food, quantity in zip(foods, quantities):
    result = data[
        data["Food_Name"].str.lower() == food.lower()
    ]
    if not result.empty:
        calories_per_100g = result.iloc[0]["Calories_per_100g_kcal"]
        calorie = (calories_per_100g * quantity) / 100
        calories.append(calorie)
        print(
            food,
            "=",
            quantity,
            "g =",
            calorie,
            "kcal"
        )
```

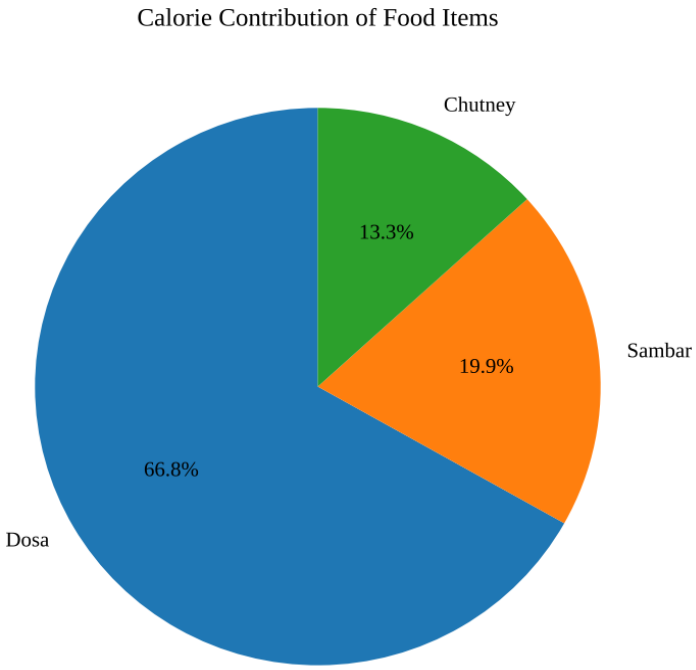
OUTPUT

Dosa = 150 g = 252.0 kcal
Sambar = 100 g = 75.0 kcal
Chutney = 50 g = 50.0 kcal

Pie Chart

```
plt.figure(figsize=(7, 7))
plt.pie(
    calories,
    labels=foods,
    autopct="%1.1f%%",
    startangle=90
)
plt.title("Calorie Contribution of Food Items")
plt.show()
```

OUTPUT



Total Calorie Report

```
total_calories = sum(calories)
average_calories = total_calories / len(calories)
highest_index = calories.index(max(calories))

print("========== FOOD CALORIE REPORT ==========")
for food, calorie in zip(foods, calories):
    print(food, ":", calorie, "kcal")

print("----------------------------------------")
print("Total Calories :", total_calories, "kcal")
print("Average Calories :", round(average_calories, 2), "kcal")
print("Highest Calorie Food :", foods[highest_index])
print("Highest Calories :", calories[highest_index], "kcal")
```

OUTPUT

========== FOOD CALORIE REPORT ==========
Dosa : 252.0 kcal
Sambar : 75.0 kcal
Chutney : 50.0 kcal

Total Calories : 377.0 kcal
Average Calories : 125.67 kcal
Highest Calorie Food : Dosa
Highest Calories : 252.0 kcal